Counting without missing anything

Kevin Hu
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L2-CO1 Level 2 Intermediate · Combinatorics

Count arrangements and selections reliably with the multiplication principle, factorials, choosing, complementary counting and well-organised casework.
By the end you can count arrangements and selections with the multiplication principle, factorials and $\binom{n}{r}$; handle conditions by filling restricted places first, gluing blocks, using gaps or counting the complement; organise casework and systematic lists so that nothing is missed or counted twice.
Prerequisites: Listing and counting systematically (L1-CO1). **Contests:** IMC, Grey Kangaroo, Pink Kangaroo, Cayley, Hamilton, JMO, AMC 8, AMC 10. **Time:** about 75 minutes.

1 Lesson

How many of the whole numbers from 1 to 999 contain at least one digit 5? One might split by position. A 5 in the units place: 100 numbers. A 5 in the tens place: another 100. A 5 in the hundreds place: another 100. This gives 300, which is wrong, since 555 has been counted three times and 505 twice. The overcount can be corrected, but the following method is shorter.

Write every number from 0 to 999 as a three-digit string by adding leading zeros, so 7 becomes 007. Now each number is a choice of three digits, and there are $10 \times 10 \times 10 = 1000$ strings. The strings with *no* 5 have 9 options in each place, so there are $9^3 = 729$ of them, and one of these is 000, which is not in our range. So 728 of the numbers from 1 to 999 avoid the digit 5, and the answer is $999 - 728 = 271$.

We used two ideas: describe each object as a sequence of choices, and count what you do not want when that is easier. This unit develops both, with the aim of counting every object exactly once.

1.1 The multiplication principle

Multiplication principle.
Suppose every object you are counting is built by making a first choice in a ways, then a second choice in b ways, then a third in c ways, and so on, where the *number* of options at each stage does not depend on what was chosen earlier, and different sequences of choices give different objects. Then there are $a \times b \times c \times \cdots$ objects.

To see why, draw the choices as a tree. The first stage gives a branches, each of those splits into b, so there are ab branches after two stages, and so on. Each complete path is one object, and each object is one path.

Both conditions in the box matter. The second (different choices give different objects) fails if you count pairs of socks as "first sock, then second sock", since each pair arises twice. The first fails

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often with digits. Count the four-digit odd numbers with all digits different. Going left to right, the first digit has 9 options (not 0), the second 9 and the third 8, but the number of options for the last digit (odd and unused) depends on how many odd digits have already been used. The remedy is to **make the most restricted choice first**. The units digit has 5 options, then the leading digit has 8 (not 0 and not the units digit), then the hundreds digit has 8 and the tens digit 7, giving $5 \times 8 \times 8 \times 7 = 2240$.

Alongside it sits the **addition principle**: if the objects split into cases with no overlap, count each case and add. Most problems use both. Split into cases, then multiply inside each case.

1.2 Arrangements and choices

The number of ways to arrange n different objects in a row is $n! = n(n-1)\cdots 2 \cdot 1$, since there are n options for the first place, $n-1$ for the second, and so on. If you only fill r places, you get $n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!}$ ordered selections.

Often order does not matter: a team, a hand of cards, a set of squares to shade. Each unordered set of r objects can be put in order in $r!$ ways, so among the ordered selections every set has been counted exactly $r!$ times. Dividing by $r!$ gives the following formula.

> The number of ways to choose r objects from n different objects, when order does not matter, is
>
> $$\binom{n}{r} = \frac{n!}{r!\,(n-r)!}.$$
>
> Also $\binom{n}{r} = \binom{n}{n-r}$, since choosing the r objects you take is the same as choosing the $n-r$ you leave behind.

The same "label, count, then divide" idea handles repeated letters. To arrange the letters of LEVEL, pretend they are L_1, E_1, V, E_2, L_2. There are $5! = 120$ arrangements of these, and each actual word appears $2! \times 2! = 4$ times (swap the Ls, swap the Es), so there are $120/4 = 30$ words. Dividing is only safe when every object is overcounted the same number of times, so check this each time.

Know a few values by heart: $\binom{n}{2} = \frac{n(n-1)}{2}$, $\binom{6}{3} = 20$ and $\binom{8}{4} = 70$.

1.3 Restrictions

Most contest counting questions are a standard count plus a condition. At this level we have four strategies for the condition.

- **Restricted places first.** As with the odd numbers above, fill the places with the fewest options before the free ones.
- **Glue.** If certain objects must be together, treat them as one block, arrange the blocks, then multiply by the number of orders inside the block.
- **Gaps.** If certain objects must be apart, arrange everything else first. Then k objects in a row create $k+1$ gaps (counting both ends), and putting the restricted objects into different gaps keeps them apart.
- **Complement.** If the condition says "at least one" or "not all", count everything and subtract the objects that fail.

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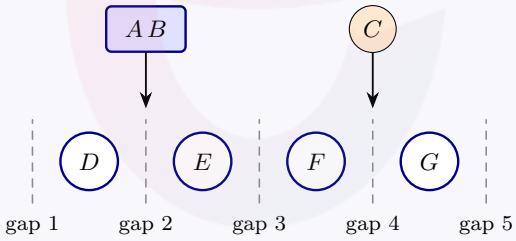
Finally, **systematic listing** is a valid method, not a last resort. When the total is small, say under 30, list in a fixed order (alphabetical, or by the first entry) so that you can see the list is complete. An organised list is a proof that the count is right; a disorganised one is not.

Example 1.
How many three-digit even numbers greater than 500 have all three digits different?

Solution.
The restricted places are the first digit (5, 6, 7, 8 or 9) and the last digit (0, 2, 4, 6 or 8). If we fill them first, the first digit has 5 options, but the number of even digits left for the last place depends on whether the first digit was even. Hence we split into cases.
If the first digit is odd (5, 7 or 9), the last digit can be any of the 5 even digits, and the middle digit is any of the 8 digits not yet used, giving $3 \times 5 \times 8 = 120$. If the first digit is even (6 or 8), the last digit has only 4 options, and the middle again has 8, giving $2 \times 4 \times 8 = 64$. The cases do not overlap, so the total is $120 + 64 = 184$.
Note that the middle digit was left until last on purpose, since it has 8 options in every case.

Example 2.
Seven people, including Asha, Ben and Cara, sit in a row of seven chairs. Asha and Ben insist on sitting next to each other, and neither of them will sit next to Cara. In how many ways can the seven sit?

Solution.
"Next to each other" suggests glue, and "not next to" suggests gaps. Glue Asha and Ben into one block, which can be AB or BA, so there are 2 internal orders. Now seat the four other people (call them D, E, F and G) in $4! = 24$ ways. They create 5 gaps: before the first, between neighbours, and after the last.



The block and Cara must go into *different* gaps, since sharing a gap would put Cara beside the block. That gives $5 \times 4 = 20$ ordered choices of gaps. Every valid seating arises exactly once, so the total is $2 \times 24 \times 20 = 960$.
We check by a second method. With Asha and Ben glued there are $2 \times 6! = 1440$ seatings in all. In the bad ones Cara sits at one end of the block, forming a block of three: CAB, CBA, ABC or BAC. There are 4 such triples and $5! = 120$ ways to arrange the five units, so 480 bad seatings, and indeed $1440 - 480 = 960$. Agreement between two independent methods is a reliable check.

Example 3.
A committee of 4 is chosen from 6 boys and 5 girls. In how many ways can this be done if the committee must include at least one boy and at least one girl?

Solution.

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A tempting idea is to guarantee the condition directly: choose a boy (6 ways), choose a girl (5 ways), then choose any 2 of the remaining 9 pupils ($\binom{9}{2} = 36$ ways), giving $6 \times 5 \times 36 = 1080$. That is more than $\binom{11}{4} = 330$, the number of committees with no condition at all, so it must be wrong. A committee with two boys and two girls is produced once for each choice of "guaranteed" boy and "guaranteed" girl, so it is counted 4 times, while a committee with three of one sex is counted 3 times. The overcount is not uniform, so dividing by a constant cannot correct it.
The condition fails only when the committee is all boys or all girls, so count the complement instead. There are $\binom{6}{4} = 15$ all-boy committees and $\binom{5}{4} = 5$ all-girl ones, so the answer is $330 - 15 - 5 = 310$.

Example 4.
A digital clock shows the time in the 24-hour format HH:MM, from 00:00 to 23:59. For how many minutes of the day are the four digits on display all different?

Solution.
A left-to-right count fails at once: after the hour is fixed, the number of options for the first minute digit (0 to 5) depends on whether the hour digits lie among 0 to 5. So split into cases according to the hour, and inside each case use the multiplication principle.
Fix the hour first, with its two digits different. The first hour digit is always 0, 1 or 2, so it always uses up one of the six digits 0 to 5. If the second hour digit is also at most 5, the first minute digit has $6 - 2 = 4$ options; if the second hour digit is 6 or more, it has $6 - 1 = 5$. Either way the last digit then has $10 - 3 = 7$ options.
Now count hours. With first digit 0, the second digit is 1 to 9: five of these (1 to 5) are at most 5 and four (6 to 9) are bigger, giving $5 \times 4 \times 7 + 4 \times 5 \times 7 = 280$. First digit 1 works the same way, with second digit 0 or 2 to 9, giving another 280. First digit 2 allows only 20, 21 and 23, all with second digit at most 5, giving $3 \times 4 \times 7 = 84$. The total is $280 + 280 + 84 = 644$.
Note that we split only on the one feature that changed the count (whether the second hour digit is at most 5).

Common traps.

- Multiplying when the number of options at a stage depends on earlier choices. Leading zeros are the classic case.
- Counting ordered selections when the question wants unordered ones, or the reverse. Ask: would swapping two chosen objects give a different answer to the question?
- "Guarantee one, then choose the rest freely." This almost always overcounts, as in Example 3.
- Forgetting the internal orders of a glued block, or forgetting the two end gaps.
- Cases that overlap, such as "the first digit is 5" and "the last digit is 5". Cases must split the objects with no overlap.
- Fence-post errors: there are $300 - 100 + 1 = 201$ integers from 100 to 300 inclusive, not 200.

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1.4 How to spot this in a contest

Almost every question beginning "how many ways" or "how many numbers" belongs to this unit. Before computing anything, decide what a single object looks like and write one down; this alone fixes most order-versus-no-order confusion. Then scan the conditions. "Different" or "distinct" suggests the multiplication principle with shrinking options, "next to" suggests glue, "not next to" suggests gaps, and "at least one" suggests the complement.

When the numbers are small, count a smaller version by hand and test your method on it. If your method says there are 6 ways to seat three people in a row with two particular people apart, and listing shows only 2, you have found the mistake. On the IMC and the Kangaroos the options are often close together, so an overcount by a few will usually lead to a wrong option.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ A password consists of two different letters chosen from A, B, C, D, E and F, followed by three digits, the last of which is odd. How many different passwords are possible?

Problem 2 ⋆⋆ How many three-digit numbers have at least two digits that are equal?

Problem 3 ⋆⋆ In how many of the different arrangements of the eight letters of the word TEETOTAL are the three Ts next to each other?

Problem 4 ⋆⋆⋆ Eight chairs stand in a row. Three people, Dev, Eli and Fay, sit down so that no two of them are in neighbouring chairs. In how many ways can this happen?

Problem 5 ⋆⋆⋆ A 5 by 5 grid of unit squares is drawn. How many rectangles whose sides lie along the lines of the grid are *not* squares?

Problem 6 ⋆⋆⋆⋆ How many four-digit numbers contain the block 12, that is, a digit 1 immediately followed by a digit 2? (For example, 1203, 7812 and 1212 do, but 2134 does not.)

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **UKMT Pink Kangaroo 2019 Question 5** ⋆
   One choice, then a second choice that must differ from the first
2. **UKMT Grey Kangaroo 2022 Question 6** ⋆
   Fill the digits one at a time; the hundreds digit has the fewest options
3. **AMC 10A 2017 Problem 8** ⋆⋆
   Count pairs of people; decide which kinds of pair shake hands
4. **UKMT IMC 2019 Question 4** ⋆⋆
   Systematic list, organised by the first digit of the hour

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5. **AMC 10A 2015 Problem 10** ⋆⋆
   Only 24 arrangements: list them in an organised way, or ask where b can go

6. **AMC 10A 2006 Problem 21** ⋆⋆
   At least one: count the complement

7. **UKMT IMC 2015 Question 20** ⋆⋆
   Multiplication principle; does the last character involve any choice?

8. **AMC 10A 2018 Problem 4** ⋆⋆
   Choose the three periods first, then assign the courses to them

9. **UKMT IMC 2017 Question 23** ⋆⋆⋆
   Seat the sisters first and use the gaps, or count the complement

10. **AMC 8 2011 Problem 23** ⋆⋆⋆
    Split into cases by the units digit, then fill the restricted places first

11. **UKMT Pink Kangaroo 2022 Question 17** ⋆⋆⋆
    Some removals are forced to happen in a fixed order; choose positions

12. **UKMT IMC 2015 Question 23** ⋆⋆⋆
    Count the arrangements that come before it, one letter at a time

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4 Hints

1. Build a password one character at a time. Does any choice affect the number of options for a later one?
2. Count the three-digit numbers whose digits are all different, and work from there.
3. Glue the three Ts into a single block. Which of the remaining letters are identical?
4. Think of the five empty chairs as already placed. Where can the three people go?
5. A rectangle is fixed by its two vertical sides and its two horizontal sides. How many grid lines are there in each direction?
6. Count the numbers with the block in each of the three possible positions. Then ask which numbers you have counted twice, and remember that a four-digit number cannot start with 0.

5 Solutions

Problem 1

A password consists of two different letters chosen from A, B, C, D, E and F, followed by three digits, the last of which is odd. How many different passwords are possible?

The first letter can be any of the 6 letters and the second any of the other 5, giving $6 \times 5 = 30$ ways to choose the letters in order. The first two digits can each be any of the 10 digits, and the last digit must be one of the 5 odd digits, giving $10 \times 10 \times 5 = 500$ ways. By the multiplication principle there are $30 \times 500 = 15000$ passwords.

Answer: 15000

Problem 2

How many three-digit numbers have at least two digits that are equal?

There are $999 - 100 + 1 = 900$ three-digit numbers. Those with all three digits different are counted by choosing the first digit (9 options, not 0), then the second (9 options, any digit except the first), then the third (8 options), which gives $9 \times 9 \times 8 = 648$. Every other three-digit number has at least two equal digits, so the answer is $900 - 648 = 252$.

Answer: 252

Problem 3

In how many of the different arrangements of the eight letters of the word TEETOTAL are the three Ts next to each other?

Since the Ts are identical, the three of them together form a single block TTT with only one internal order. We are then arranging six objects: the block, E, E, O, A and L. If the two Es were different there would be $6! = 720$ arrangements, but swapping the Es gives the same word, so each word has been counted twice. The answer is $720/2 = 360$.

Answer: 360

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Problem 4

Eight chairs stand in a row. Three people, Dev, Eli and Fay, sit down so that no two of them are in neighbouring chairs. In how many ways can this happen?

Read a seating from left to right as a row of 5 empty chairs and 3 people. The 5 empty chairs create 6 gaps (one at each end and four between neighbouring empty chairs). No two people are neighbours exactly when the three people lie in three different gaps, with at most one person per gap. So a seating is determined by choosing which 3 of the 6 gaps are used, which can be done in $\binom{6}{3} = 20$ ways, and then deciding which person goes in which chosen gap, which can be done in $3! = 6$ ways. Each seating arises exactly once this way, so there are $20 \times 6 = 120$ seatings.

Answer: 120

Problem 5

A 5 by 5 grid of unit squares is drawn. How many rectangles whose sides lie along the lines of the grid are not *squares?*

The grid has 6 vertical lines and 6 horizontal lines. A rectangle is determined by choosing two of the vertical lines for its left and right sides and two of the horizontal lines for its top and bottom, and different choices give different rectangles. So there are $\binom{6}{2} \times \binom{6}{2} = 15 \times 15 = 225$ rectangles.

A k by k square can be placed with its bottom left corner in $(6-k)^2$ positions, so the number of squares is $5^2 + 4^2 + 3^2 + 2^2 + 1^2 = 55$. The number of rectangles that are not squares is $225 - 55 = 170$.

Answer: 170

Problem 6

How many four-digit numbers contain the block 12, that is, a digit 1 immediately followed by a digit 2? (For example, 1203, 7812 and 1212 do, but 2134 does not.)

The block can start in the first, second or third place. Let X, Y and Z be the sets of four-digit numbers with the block in places 1 and 2, in places 2 and 3, and in places 3 and 4 respectively. We want the number of numbers in at least one of these sets.

Numbers in X have the form $\overline{12cd}$, with 10 choices for each of c and d, so there are 100 of them. Numbers in Y have the form $\overline{a12d}$, where the first digit a cannot be 0, so there are $9 \times 10 = 90$. Numbers in Z have the form $\overline{ab12}$, again with $a \neq 0$, so there are $9 \times 10 = 90$. The total $100 + 90 + 90 = 280$ counts a number twice if it lies in two of the sets.

A number cannot lie in both X and Y, since its second digit would have to be both 2 and 1. For the same reason no number lies in both Y and Z. A number in both X and Z is $\overline{1212}$, and there is only one. So exactly one number has been counted twice and none three times, and the answer is $280 - 1 = 279$.

Answer: 279